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# Utilizing CMIP6-SSP scenarios with the VIC model to enhance agricultural and ecological water consumption predictions and deficit assessments in arid regions

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ABSTRACT

Essential for economic development and ecological restoration in arid regions, water resources are currently facing unprecedented scarcity. Although future changes in surface water resources have been extensively examined, there has been limited focus on the balance between agricultural and ecological water consumption and water deficits, particularly in arid basins. A novel approach to estimating agricultural and ecological water consumption was introduced in this study using a physical process model (Variable Infiltration Capacity model, VIC) incorporating the latest Coupled Model Intercomparison Project Phase 6 (CMIP6) Shared Socioeconomic Pathway (SSPs) scenarios. Water consumption trends and deficits were analysed using historical data (1961–2014) and projected under the SSP126, SSP245, SSP370, and SSP585 scenarios for 2015–2100 in the Lake Ebinur basin. There was a significant increase in agricultural and ecological water consumption along with a growing water supply deficit, particularly under the SSP245 and SSP585 scenarios. Agricultural water consumption is projected to increase by 8.00% to 43.13%, ecological water consumption is expected to rise by 13.31% to 49.11%, and the water supply deficit is projected to increase by 45% to 113% relative to the baseline period. The average annual mean error of raw meteorological variables was reduced by 71.66% after applying bias correction, leading to an improvement of approximately 86.79% in the simulation accuracy of the VIC model compared with the uncorrected scenario. An increase in precipitation ranging from 4.00% to 33.56%, a maximum temperature increase of 230.88%, and a decrease in wind speed of 6.45% were projected for the mid-to-late 21st century under the SSP585 scenario. The water supply deficit was estimated to increase under the SSP245 scenario, with deficits projected to reach 1.35 billion $m^3$ per year in the medium term and 1.59 billion $m^3$ per year in the late term. Given the projected increase in agricultural and ecological water consumption and the growing future water supply deficit, quantifying these changes can provide critical insights into water resource management and sustainable development in arid regions.

## 1. Introduction

Global water shortages are worsening, with projections showing that 2.37 billion people could be affected by 2050, This is particularly the case in regions like China's northwest, where increasing water stress, coupled with the growing competition between ecological and socio-economic water demands, is further depleting available water resources (Liu et al., 2024). The escalation of water scarcity has led to environmental degradation, desertification, declining food production, and soil salinisation. This poses considerable challenges to arid regions, such as the Aral Sea, Tarim River, and Black River Basins, where ecosystem health and socioeconomic development are increasingly threatened (Azzam et al., 2022). Balancing the increasing conflict between socioeconomic and ecological water demands is essential for mitigating growing water scarcity and ensuring the long-term sustainability of ecosystems and livelihoods in arid regions.

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0168-1699/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Evapotranspiration is the primary factor considered in water consumption estimation studies because it represents the main form of water consumption in both ecological and agricultural systems. Although hydrological and crop growth models are widely used to simulate evapotranspiration, the acquisition of large-scale temporally continuous input data poses a substantial challenge. The complexity of model parameters and structures often leads to inaccurate simulation results. When driven by remote sensing data, hydrological models can simulate evapotranspiration over long spatiotemporal scales. This can provide high resolution and enhanced interpretability of the underlying physical processes, offering advantages over contemporary machine learning models that often lack such physical transparency (Jiang et al., 2024). Future human activities and climate change are expected to alter evapotranspiration, disrupting the balance of water resource consumption in ecological and agricultural systems, particularly in arid regions. Consequently, there is an urgent need to develop a high-precision technical framework to predict the changes in water resource consumption across different systems.

Using hydrological models in combination with Global Climate Model outputs for water resource simulation and prediction is an effective approach for implementing this water consumption estimation framework. With the release of the latest Intergovernmental Panel on Climate Change (IPCC) Coupled Model Intercomparison Project Phase 6 (CMIP6), which incorporates updated carbon emission scenarios and socioeconomic impacts through shared socioeconomic pathways (SSPs), CMIP6 climate projections offer more plausible outcomes than previously used representative concentration pathways (RCPs) (Xu et al., 2023). Li et al. (2022) combined an improved SWAT model with CMIP5 and CMIP6 GCM projections, finding that future climate change could increase winter wheat and summer maize yields by up to 18.9 % and 16.7 %, respectively, while altering evapotranspiration and irrigation demands. Jamnani et al. (2024) used the WEAP and MODFLOW models with CMIP6 climate data to simulate future water resources. They found that under SSP8.5, a 15 % decrease in precipitation could reduce reservoir inflow by 25 % and groundwater levels by 1.2 m annually, causing a 65 million $m^3$ shortfall in agricultural water demand. While most prior studies have focused on estimating traditional hydrological fluxes, relatively few have examined the future tradeoffs between agricultural and ecological water consumption in arid regions, particularly using physical process models with global climate projections. Therefore, it is necessary to predict future trends in agricultural and ecological water consumption in arid inland regions and to quantify the water supply–demand deficit under different scenarios. This is crucial for the optimal allocation and efficient use of water resources.

In the Lake Ebinur Basin, a substantial decrease in water surface area was observed from 743.37 $km^2$ to 336.68 $km^2$ from 2016 to 2023. This reflects the severe water scarcity and environmental degradation in this arid region of northwestern China. Here, human activities, such as cropland expansion and overgrazing, increase the frequency of environmental challenges, including desertification and soil salinisation (Shi et al., 2024). The contributions of climate change and human activity to runoff variation have been 46.87 % and 53.13 %, respectively, over the last 50 years (Wang et al., 2021). Zeng et al. (2019) showed that restoring the lake to 522 $km^2$ required an increase in annual inflow by $602.6 \times 10^6$ $m^3$ and a reduction in human water consumption by $320.3 \times 10^6$ $m^3$. This emphasises the delicate balance between ecological restoration and human water usage in arid regions. Understanding the responses of water resources to future climate and human activity changes is essential for ensuring sustainable water use at the basin scale in arid inland regions. This highlights the need for integrated tools, such as physical hydrological models coupled with climate projections, to simulate the spatiotemporal dynamics of water consumption.

The Lake Ebinur basin was selected as the case study because of its ecological degradation resulting from water scarcity. Over the last 50 years, the expansion of artificial oases has resulted in severe water scarcity, primarily driven by human activities. At this site, the impacts of future climate change and human activities on agricultural and ecological water consumption, along with the associated water deficits, were investigated. The specific objectives of this study were to (1) apply quantile delta mapping preprocessing to enhance the accuracy of CMIP6 climate model projections, (2) use a VIC model for hydrological simulations and uncertainty analysis, and (3) implement the proposed framework to estimate agricultural and ecological water consumption and assess the water supply–demand deficit under various future climate scenarios. The findings can contribute to improving sustainable management of water resources in arid regions.

## 2. Materials and methods

### 2.1. Study area

The Lake Ebinur Basin is located in the southwestern part of the Junggar Basin in Xinjiang, China (43°38′-45°52′N, 79°53′-85°02′E), which has important national-level nature conservation areas (Fig. 1). The study area has a typical continental arid climate with substantial annual temperature and wind speed variations, low precipitation, and high potential evapotranspiration. The area is drained by the Jinghe and Bortala rivers, which flow from the mountains into the inland basin and converge to form the tailrace of Lake Ebinur. Cotton, wheat, and corn are the dominant crops in the basin. Meanwhile, the natural vegetation consists mainly of desert species, such as camel thorn, pokeweed, and red willow, with soils classified into three main categories, namely, Entisols, Aridosols, and Mollisols. The natural environment of the basin has been deteriorating due to human activities and the impacts of climate change. The lake surface area has substantially reduced, with intense competition between socioeconomic and ecological water use. This has made water scarcity an urgent issue for the local community.

### 2.2. Data

Daily-scale meteorological variables were used to drive the VIC model, including precipitation, maximum temperature, minimum temperature, and wind speed. These were sourced from the China Meteorological Administration (CMA, https://data.cma.cn/) for 1959 to 2014. A thin-plate spline interpolation method based on climatic means was used, with the main advantage that the use of a well-developed climatic background field effectively controlled errors that were difficult to correct during the interpolation process. This could provide a more accurate climate estimate with climatic averages sourced from WorldClim (Keller and Borkowski, 2019) (https://www.worldclim.org/data). Soil and vegetation library parameters were obtained from the VICGlobal product data (https://doi.org/10.5281/zenodo.4059993), which have been well-validated (Schaperow et al., 2021). The land-cover data were obtained from the MODIS satellite product MCD12Q1, averaged from 2000 to 2020, processed, and downloaded from the Google Earth Engine. Topographic data, including digital elevation modelling, slope, drainage direction, flow accumulation, and stream networks, were sourced from HydroSHEDS (Gong et al., 2011). The measured runoff data were acquired from three hydrological stations of local water agencies over an extended period. Meanwhile, the evapotranspiration data were obtained from meteorological station observations, which have been assessed for their quality (Wang et al., 2021).

### 2.3. Methods

The methodology progressed through three distinct phases. This included downscaling and pre-processing Global Climate Model (GCM) data, followed by hydrological modelling to understand water dynamics. This culminated in the calculation of water consumption with a specific focus on ecological and agricultural needs. Through these steps, the projected changes in meteorological variables and their impact on water consumption were analysed, addressing the potential disparities

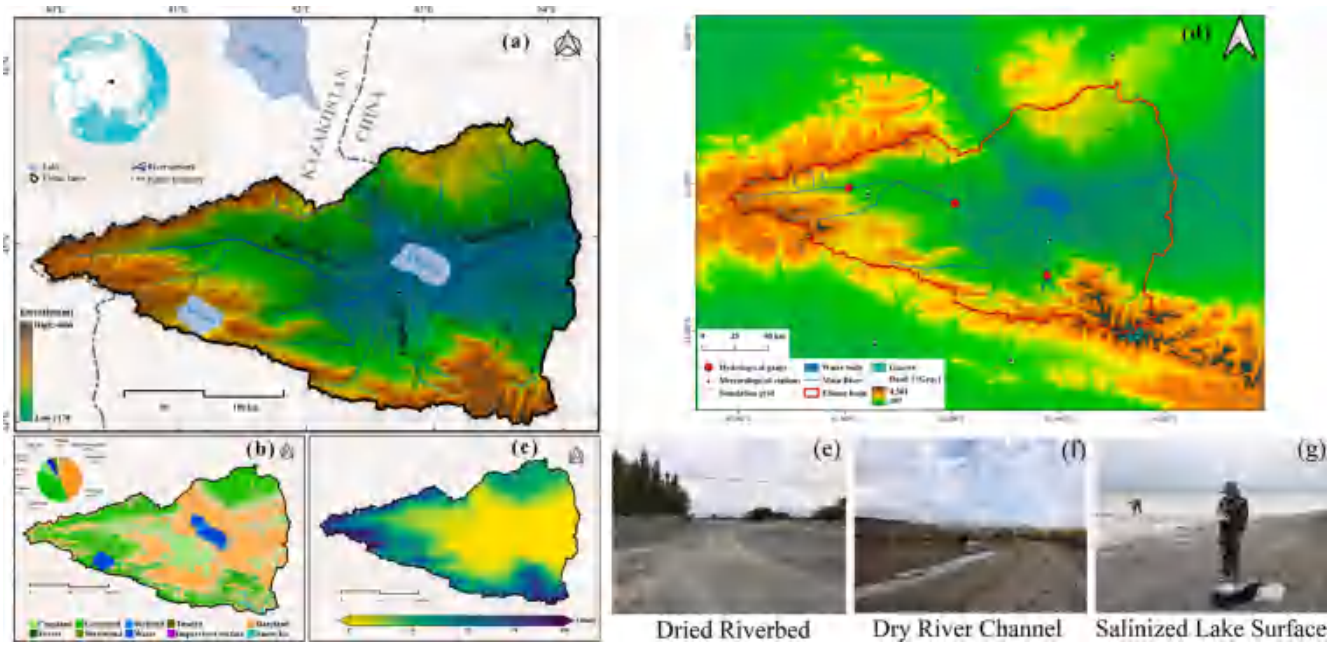


**Fig. 1.** Study area. (a) Geographical location and topography of the Ebinur Lake Basin. (b) Spatial distribution and proportion of major land cover types (land cover data sourced from http://data.ess.tsinghua.edu.cn/). (c) Spatial distribution of annual average precipitation (data sourced from https://www.worldclim.org/data/worldclim21.html). (d) Distribution of simulation grids, meteorological and hydrological stations, and glaciers (a total of 1,794 simulation grids, glacier data sourced from http://westdc.westgis.ac.cn). (e) Dried Riverbed, (f) Dry River Channel, (g) Salinized Lake Surface.

between water supply and demand under various future scenarios. Well-founded recommendations could then be provided based on these insights (Fig. 2).

### 2.3.1. VIC model

A physically based, large-scale distributed hydrologic model, the VIC model was used to calculate the energy and moisture budgets for each grid cell on a daily or sub-daily basis. This incorporated complex physical processes, such as snowpack, snowmelt, vegetation, and permafrost (Park and Markus, 2014). The study area was divided into several latitudinal and longitudinal grid cells in the VIC model. Each cell was further subdivided into multiple land-cover tiles representing the proportions of different land-cover types, such as grassland and cropland (Appendix Tables 5 and 6. The water fluxes were simulated for each land tile and used as a weighted average of the grid cells, with the mechanisms described above as the basis for calculating the ecological and agricultural water consumption. The VIC model has been extensively calibrated and validated in numerous basins worldwide, including in inland arid regions. It is a key tool for studying water resources, climate change, and the impacts of human activities (Feng et al., 2023).

A three-year model spin-up period was applied, with the historical study period divided into calibration (1961–1985) and validation (1986–2014) phases.

### 2.3.2. Framework for the calculation of agricultural and ecological water consumption

Simulated evapotranspiration (ET) estimates were used as proxies for agricultural and ecological water consumption in agricultural and natural land classes, respectively. From a water cycle perspective, ET is considered part of the natural water cycle. Here, water is absorbed by vegetation through the root system and released into the atmosphere. This allows water dissipation in both agricultural and natural land types to be approximated and quantified using ET (Yang et al., 2023). The first step involved calculating the percentages of agricultural and natural land classes within the original VIC land-cover parameter grids by summing the corresponding land class data such as

agricultural land class codes 11 and 13 for each grid. The second step involved extracting the VIC simulated evapotranspiration for each grid. This was then multiplied by the land class percentages to obtain the evapotranspiration for the agricultural or natural land classes (Fig. 3).

### 2.3.3. Climate change scenarios

*2.3.3.1. Coupled model Intercomparison Project Phase 6 (CMIP6).* The latest generation of General Circulation Models (GCMs) from the Coupled Model Intercomparison Project Phase 6 (CMIP6) was used to provide climate change and anthropogenic forcing data for the hydrological model. Variations in GCM performance across the study area were attributed to differences in the representation and parameterisation of physical processes. The GCM-CanESM5 (Canadian Earth System Model, version 5) was selected as the most suitable model for the Central Asian region (Jiang et al., 2020), with CMIP6 data obtained from the World Climate Research Programme (WCRP) portal (https://esgf-node.llnl.gov/). We analysed predicted trends in temperature, precipitation, and wind speed under SSPs scenarios for both the mid- and late 21st century. However, owing to the insufficient spatial resolution of the GCM output, downscaling and cropping were required to achieve the necessary resolution for application in the hydrological model.

In the CMIP6 framework, four SSPs scenarios, namely, SSP1-2.6 (SSP126), SSP2-4.5 (SSP245), SSP3-7.0 (SSP370), and SSP5-8.5 (SSP585), were used to project the evolution of societies and their influence on climate change. Each scenario represents a distinct future pathway. SSP126 is centered on sustainable development, prioritising environmental protection and social well-being; SSP245 reflects a continuation of current trends with various mitigation strategies; SSP370 envisions a future marked by regional conflicts and diminished global cooperation, resulting in high energy use and emissions; and SSP585 outlines a 'worst-case' scenario characterised by high energy consumption, emissions, and the absence of effective climate policies.

*2.3.3.2. Bias correction methods.* The quantile delta mapping (QDM) method was used to minimise bias in climate modelling by preserving the relative changes in the quartiles of the original GCM predictions

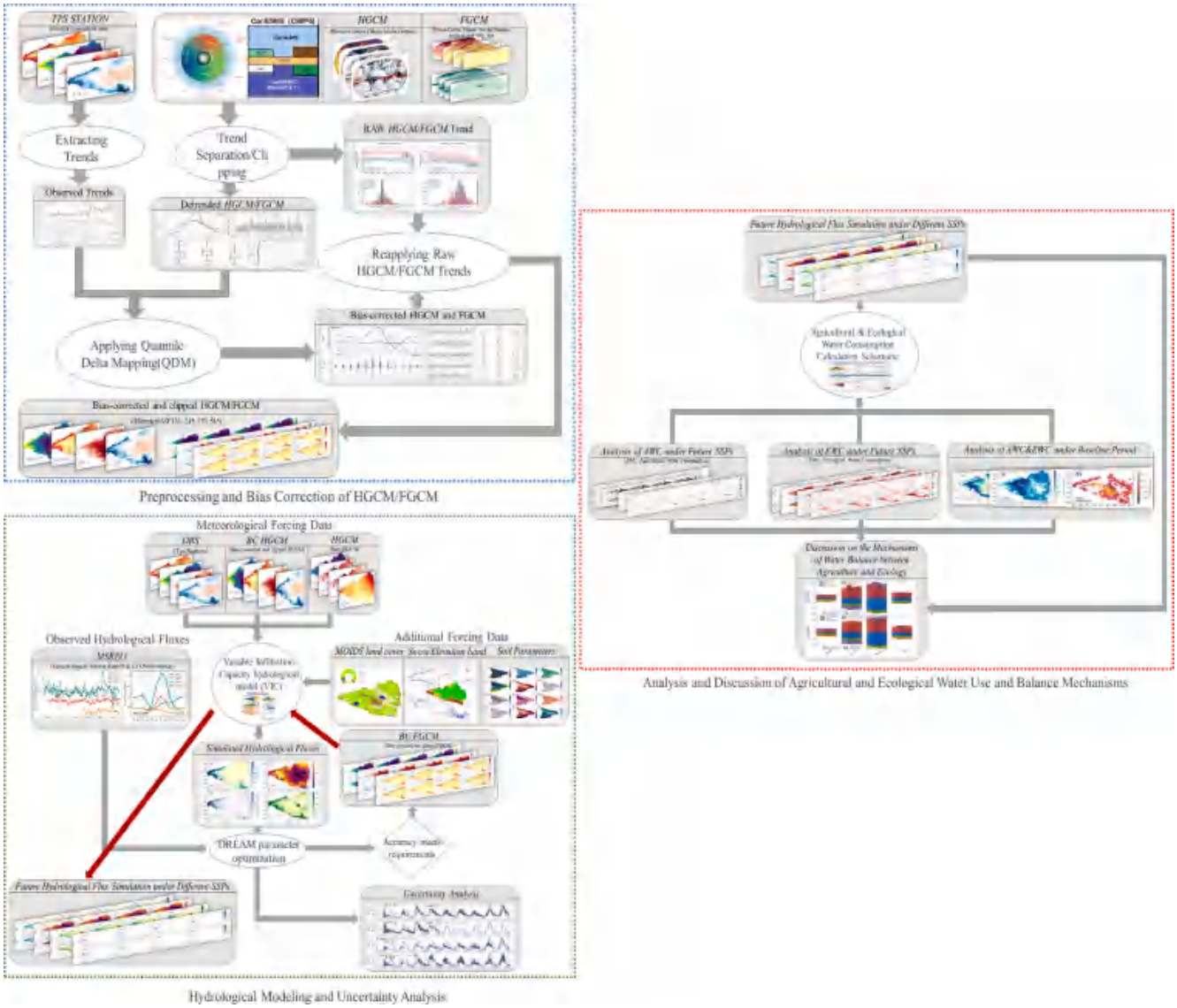


**Fig. 2.** Flowchart schematic of the methodology. (The top-left panel shows the bias correction process of GCM output data, the bottom-left panel presents the VIC hydrological modeling process, and the right panel illustrates the future assessment of water consumption and water deficits.).

while systematically adjusting for biases in these quartiles relative to observed values (Cannon et al., 2015). The concept of the QM method is to the apply the same empirical cumulative distribution functions (CDFs) to both the simulated and observed datasets. The equation is given by

$$x_{m,p}(t) = F_{o,h}^{-1}\{F_{m,h}[x_{m,p}(t)]\} \tag{1}$$

where $x_{m,p}(t)$ are the simulated data at the forecast time $t$, and $F_{m,h}$ and $F_{o,h}^{-1}$ are the CDF of the climate model's original data and the inverse CDF of the observed data, respectively.

QDM is a statistical downscaling and bias correction method in which model biases are corrected through quantile mapping while the relative changes in model predictions are preserved. Derived from both the quantile change and quantile perturbation methods, this process applies relative changes to all model quantiles, rather than just the mean of the model. The algorithm is performed in two sequential steps. First, the future outputs are detrended by a quantile, followed by bias correction through quantile mapping to align them with the observed values. Second, the relative changes in the model's quantiles are applied to the bias-corrected model outputs. The equation is given by

$$\hat{x}_{m,p}(t) = \hat{x}_{o:m,h:p}(t)\cdot\Delta_m(t) \tag{2}$$

$$\hat{x}_{o:m,h:p}(t) = \mathrm{F}_{\mathrm{o,h}}^{-1}\left[F_{m,p}^{(t)}\{x_{m,p}(t)\}\right] \tag{3}$$

$$\Delta_m(t) = \frac{x_{m,p}(t)}{F_{m,h}^{-1}\left[F_{m,p}^{(t)}\{x_{m,p}(t)\}\right]} \tag{4}$$

Where $\hat{x}_{o:m,h:p}(t)$ is the bias-corrected data for the historical period and $\Delta_m(t)$ is the relative change in the model data between the historical and future periods, with the variable used for bias correction being the meteorological station data in this study. Therefore, by multiplying the relative change $\Delta_m(t)$ by the historical bias-corrected value, the bias-corrected future prediction at time $t$ is obtained. Here, where $\Delta_m(t)$ is defined as $x_{m,p}(t)$ (simulated raw data) divided by $F_{m,h}^{-1}\left[F_{m,p}^{(t)}\{x_{m,p}(t)\}\right]$. The percentile value corresponding to the historical cumulative distribution function (CDF) of $x_{m,p}(t)$. $\Delta_m(t)$ retains the relative rate of change of the simulated raw data for each period and is considered a fundamental component of the QDM method.

The workflow for downscaling climate model outputs using the QDM

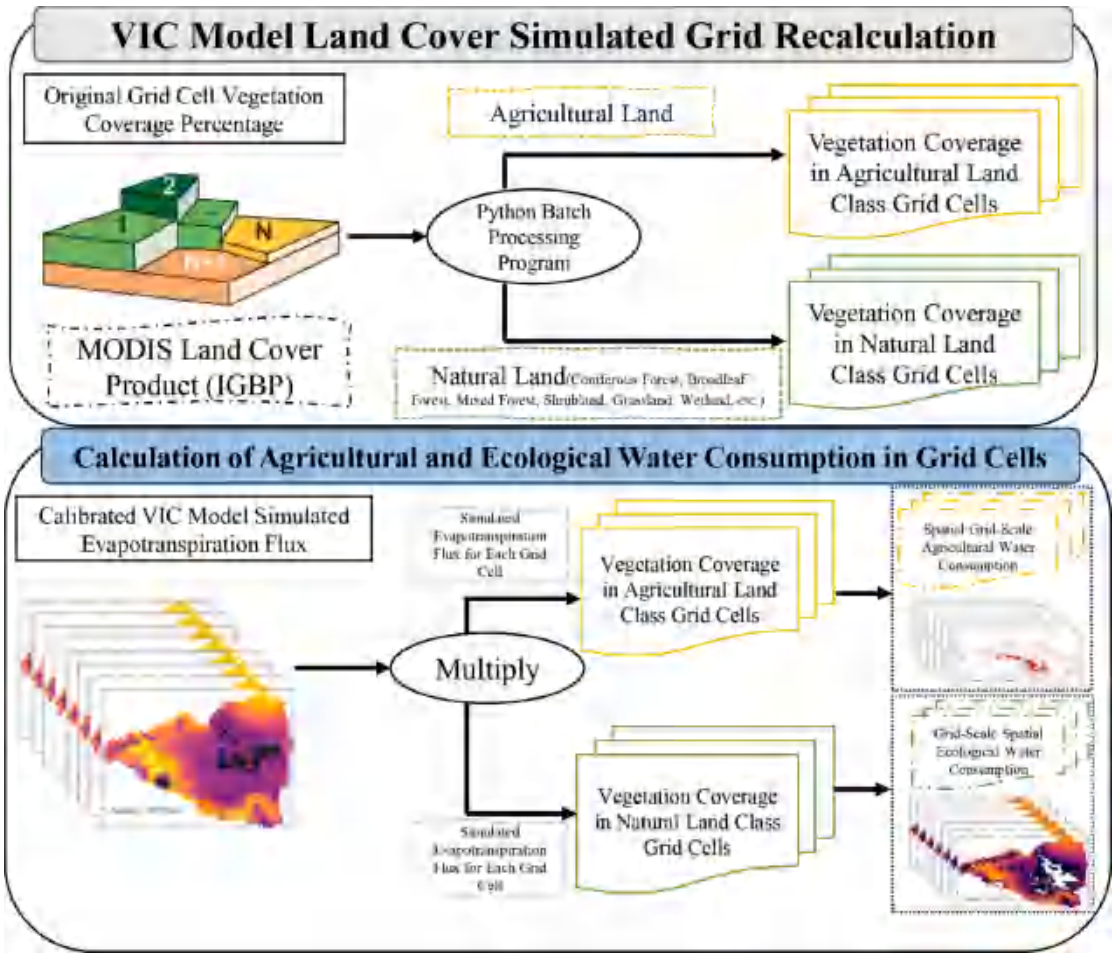


**Fig. 3.** Agricultural and ecological water consumption calculation schematic.

method is as follows: First, detrending of the climate model output is performed by quantile-based detrending, which separates long-term trends such as temperature rise from short-term variations including daily or seasonal changes, providing the foundation for subsequent bias correction. Quantile mapping was applied to all empirically detrended quantiles by adjusting the model output to match the statistical distribution of the observed data. This reduced the distributional differences between the model output and the observed data. The predicted trend was reapplied to the bias-corrected quantiles, ensuring that the model's fundamental prediction, such as temperature rise was preserved. In our study, the QDM method was applied to bias-correct the raw output from the GCM. This allowed the generation of refined meteorological simulations, both historical and future projections, based on historical observations interpolated from Thiessen Polygon Station (TPS) meteorological sites for the baseline period from 1961 to 2014.

In this study, the downscaling algorithm was evaluated based on spatial differences, CDF curves, interannual and within-year time-series differences, density scatter plots, and statistical indicators (BIAS, $R^2$, RMSE, and R). For the specific formulas of the statistical indicators, please refer to Section 2.3.4.

### 2.3.4. Validation metrics

Bias correction and VIC model performance assessment were performed using the following statistical metrics: NSE, RMSE, BIAS, $R^2$, and R, calculated as follows:

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{n}(O_t - S_t)^2}{\sum_{t=1}^{n}(O_t - \overline{O})^2} \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(O_t - S_t)^2} \tag{6}$$

$$\mathrm{BIAS} = \frac{1}{n}\sum_{t=1}^{n}(S_t - O_t) \tag{7}$$

$$R^2 = 1 - \frac{\sum_{t=1}^{n}(O_t - S_t)^2}{\sum_{t=1}^{n}(O_t - \overline{O})^2} \tag{8}$$

$$R = \frac{\sum_{t=1}^{n}(O_t - \overline{O})(S_t - \overline{S})}{\sqrt{\sum_{t=1}^{n}(O_t - \overline{O})^2 \sum_{t=1}^{n}(S_t - \overline{S})^2}} \tag{9}$$

where $O_t$ is the observed value at time $t$, $S_t$ is the simulated value at time $t$, $\overline{O}$ is the mean of the observed values, $\overline{S}$ is the mean of the simulated values, and $n$ is the total number of data points. The thresholds for accepting the error for NSE, RMSE, BIAS, $R^2$, and R are typically set based on the settings of most studies (Baker and Taylor, 2016), with values close to 1 for NSE, low values for RMSE and BIAS (preferably near 0), values above 0.7 for $R^2$, and values above 0.7 for R indicating acceptable model performance. They were used as statistical indicators to assess the degree of agreement, discrepancy, systematic error, goodness of fit, and correlation between the simulation and actual observations.

## 3. Results

### 3.1. Evaluation of bias correction performance

Fig. 4 shows the differences between precipitation, temperature, wind speed, and observations at meteorological stations for the historical period (1961–2014) before and after bias correction using the QDM method. After applying the QDM bias correction, the differences between the precipitation, maximum temperature, minimum temperature, and wind speed variables and the observed values were reduced from the original 45.23 mm, −1.60 °C, −4.00 °C, and 1.13 m/s per year to −1.32 mm, −0.01 °C, 0.003 °C, and 0.002 m/s per year, respectively. This has demonstrated the effectiveness of the QDM bias correction (Fig. 4a–e). The CDF curve analysis showed that the QDM-corrected data curves (green dotted line) closely matched the observed value curves (blue solid line). QDM significantly improved the differences between the raw precipitation data in the 0–15 mm range, raw maximum temperatures in the − 5 to 25 °C range, raw minimum temperatures in the −35 to 15 °C range, raw wind speeds in the 0–8 m/s range, and the observed values (Fig. 4f). The basin-wide monthly mean biases and overall bias box plots show that all variables exhibit monthly biases of approximately 0 when compared with the observed values (Fig. 4g). Higher precipitation biases were observed from May to October ranging from 1 to − 1 mm, with the basin bias peaking from June to July with an interquartile range between 0 and 20 mm (Fig. 4h). The maximum and minimum temperatures exhibited higher biases from January to April ranging from 0 to 0.5 °C, with the overall basin bias being greater from March to May and September to November (Fig. 4h). The wind speed showed less variation in bias across all months, with the overall basin bias being higher from January to March and September to December ranging from 0 to 2.5 m/s (Fig. 4h).

A comparison of meteorological observations and climate model simulations before and after bias correction during the historical period is shown in Fig. 5. Appendix Figs. 1-3 show the maximum temperature, minimum temperature, and wind speed variables. The results after the QDM bias correction were suitable for $R^2$, BIAS, RMSE, and R. The raw values for precipitation were 0.17, −45.23 mm, 153.17 mm, and 0.62, with corrected values of 0.77 mm, 1.32 mm, 79.61 mm, and 0.88 mm, respectively (Fig. 5b–c). The raw values for maximum temperature were 0.51 °C, 0.97 °C, 4.53 °C, and 0.73 °C, with corrected values of 0.92 °C, 0.02 °C, 1.34 °C, and 0.97 °C, respectively (Appendix Fig. 1b–c). The raw values for minimum temperature were 0.53 °C, 4.00 °C, 6.91 °C, and 0.13 °C, with corrected values of 0.97 °C, 0.01 °C, 0.97 °C, and 0.98 °C, respectively (Appendix Fig. 2b–c). The raw values for wind speed were 0.49 m/s, −0.66 m/s, 0.97 m/s, and 0.95 m/s, with corrected values of 0.92 m/s, 0.001 m/s, 0.36 m/s, and 0.96 m/s, respectively (Appendix Fig. 3b–c). The inter-annual time-series trend indicates that the gap between the QDM-corrected precipitation, air temperature, and wind speed and the observed values was narrowed by 24.61 %, 16.67 %, and 43.51 %, respectively, on the annual average scale of change (Fig. 5 and Appendix Fig. 1-3d–h). The gap between the variables was reduced by approximately 10–30 % on a seasonal scale (Fig. 5 and Appendix Fig. 1-3 (a)). The mean annual trends of precipitation, temperature, and wind speed observations were at 1.17 mm/year, 0.01 (0.06) °C/year and − 0.02 m/s per year, respectively. However, the correction values were 0.04 mm/year, 0.05 (0.06) °C/year, and 0.001 m/s per year, respectively. This indicated that the bias-corrected inter-annual trends were not well reflected (Fig. 5 and Appendix Fig. 1-3d–h). The corresponding intra-annual trends show that the QDM bias correction significantly reduced the overestimation of raw precipitation from

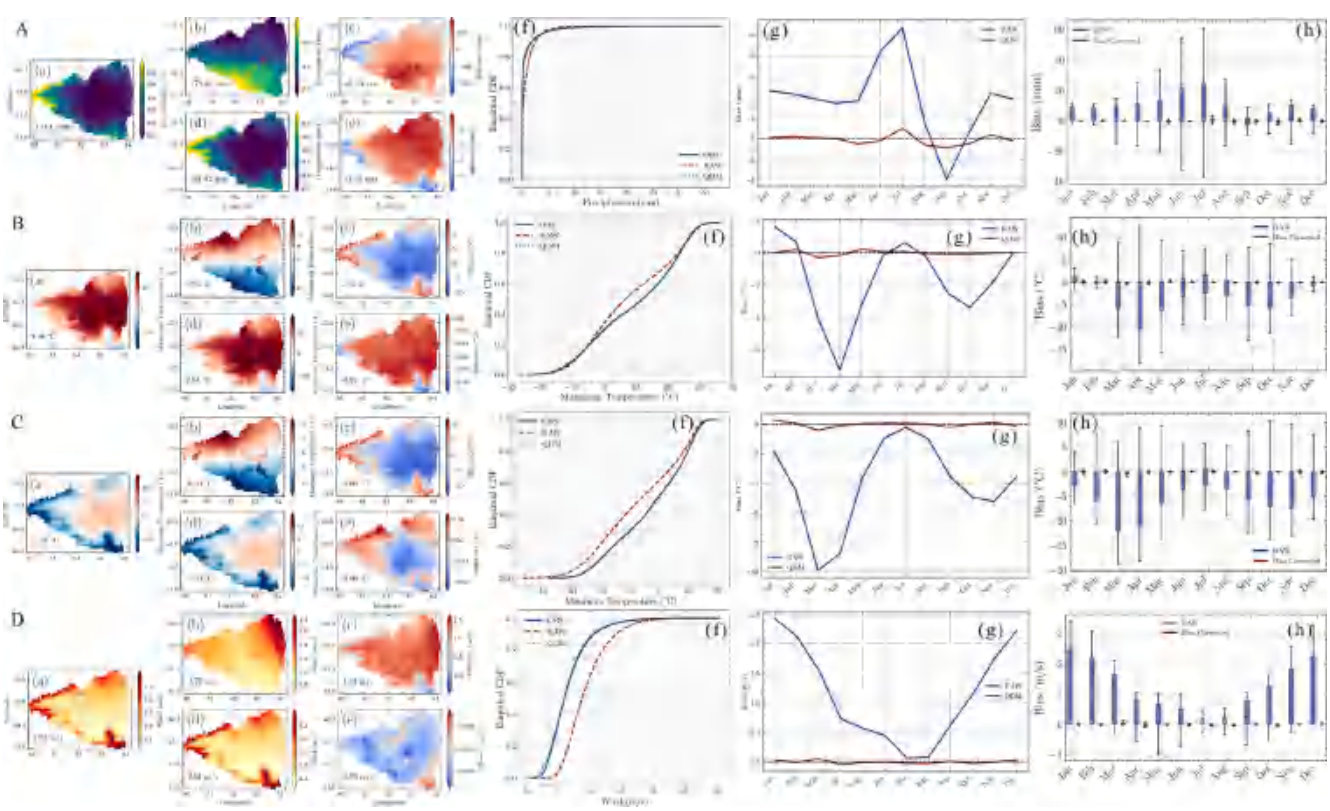


**Fig. 4.** Comparison of meteorological variables from the CMIP6-CanESM5 model outputs before and after bias correction. (Panels A, B, C, and D correspond to precipitation, maximum temperature, minimum temperature, and wind speed, respectively. Panels (a) display TPS interpolated station data, while panels (b) and (d) represent the original and QDM-corrected historical CMIP6-CanESM5 data. Panels (c) and (e) show the spatial differences between the CMIP6-CanESM5 outputs and TPS interpolated data before and after QDM correction. Panel (f) compares the Empirical Cumulative Density Functions (CDF) for TPS interpolated data (OBS), uncorrected CMIP6-CanESM5 data (RAW), and QDM-corrected data (QDM). Panels (g) and (h) highlight the monthly average biases and boxplot comparisons, respectively, between the uncorrected and QDM-corrected historical CMIP6-CanESM5 data.).

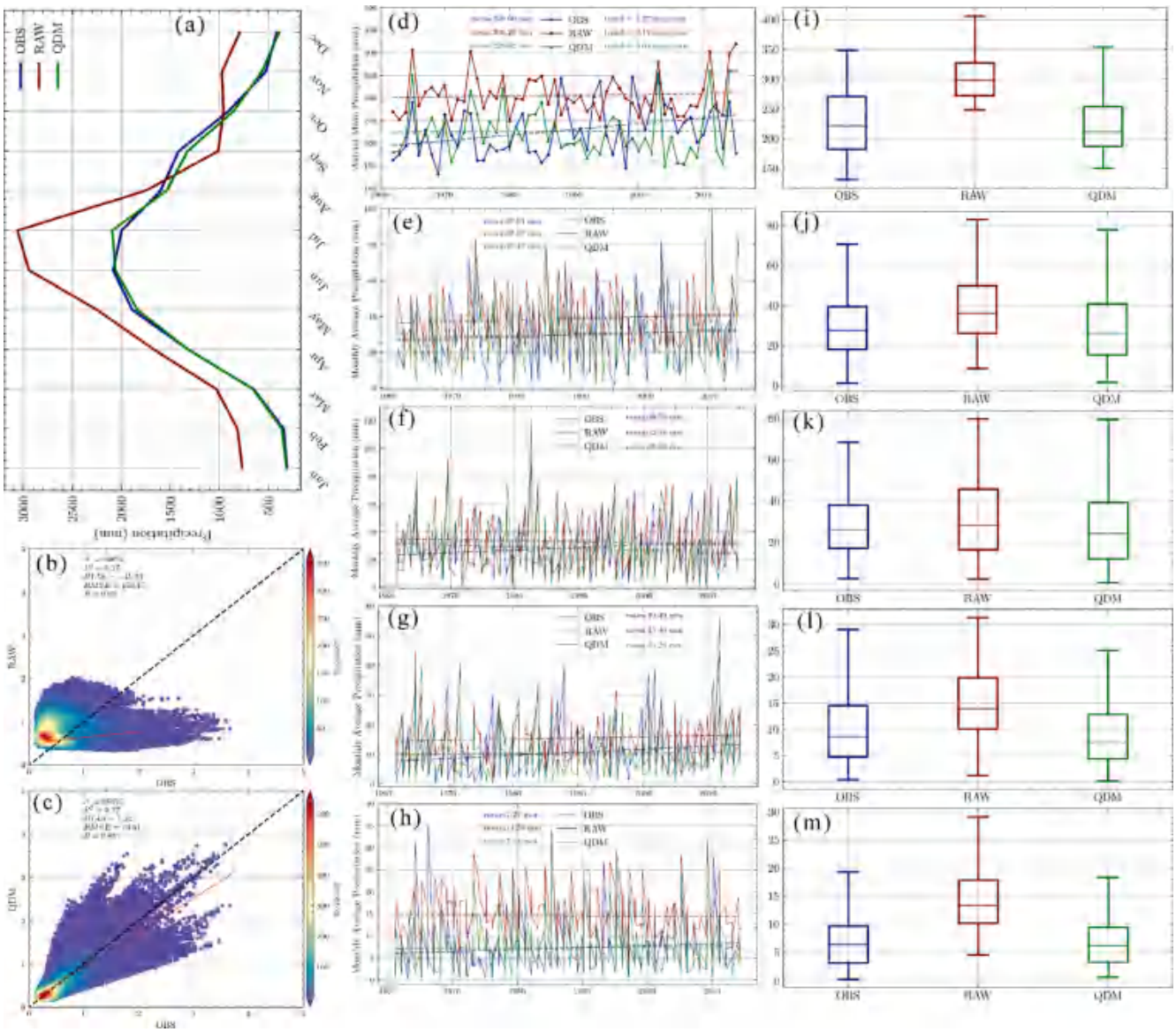


**Fig. 5.** Time series and scatter plot comparison of precipitation variables before and after bias correction in CMIP6-CanESM5 model outputs. ((a) Comparison of intra-annual time series, where OBS represents TPS-interpolated meteorological station data, RAW denotes the original historical CMIP6-CanESM5 model data, and QDM indicates the QDM bias-corrected model data. (b, c) Density scatter plots comparing the original model output, bias-corrected data, and TPS-interpolated station data. (d–m) Long-term time series changes from 1961 to 2014, where (d–h) depict annual mean and seasonal changes (spring, summer, autumn, and winter), and (i–m) show corresponding boxplots for these time periods compared to the TPS-interpolated station data.).

January to August, low raw temperatures from February to May and August to November, and contrasting raw wind speeds across all months (Fig. 5 and Appendix Fig. 1-3a). Variations in the boxplots showed that, after QDM bias correction, all variables were closer to the observed values at the median and within the quartile range (Fig. 5 and Appendix Fig. 1-3i–m).

### 3.2. Future meteorological Variable changes under different SSPs scenarios

Based on the bias-corrected future climate model output results, it was shown that, in the mid-21st century, the future mean annual precipitation is expected to increase by 4.00 %, 7.09 %, 8.96 %, and 6.41 %, respectively, compared to the baseline period under scenarios SSP126, SSP245, SSP370, and SSP585 (Table 1). By the late 21st century, it was projected to increase by 18.55 %, 24.91 %, 19.74 %, and 33.56 %, respectively. The temperature results showed an increase of up to 84.87 % in the SSP585 scenario and a minimum of 67.02 % in the SSP126 scenario in the mid-21st century, with a maximum increase of 230.88 % and a minimum of 94.69 % in the late 21st century, compared to the baseline period (Appendix Table 1). In contrast, the wind speed results showed a downward trend compared to the baseline period, with a maximum projected decrease of − 2.27 % in the SSP370 scenario and a minimum of − 1.62 % in the SSP126 scenario for the mid-21st century (Appendix Table 2). By the late 21st century, a maximum projected decrease of − 6.45 % was observed under the SSP245 scenario, with a minimum decrease of − 3.89 % under the SSP126 scenario. In terms of the differences in relative changes within the year, precipitation exhibited an upward trend from March and October to December in the mid-21st century compared to the baseline period, along with a downward trend in August and September. This trend was more pronounced under the high carbon emissions scenario (Appendix Tables 1 and 2 and Table 1). Compared with the baseline period, temperatures decreased significantly in winter and increased during the rest of the year. This

**Table 1**
Projected relative changes in precipitation by the mid-21st century (2050 s) and late 21st century (2100 s) under different SSP scenarios, relative to the baseline period (1961–2014).

| Month | Baseline | Relative change in precipitation (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | (mm) | SSP126 | | SSP245 | | SSP370 | | SSP585 | |
| | | MC | LC | MC | LC | MC | LC | MC | LC |
| Jan | 3.07 | 5.98 | 19.94 | 15.01 | 35.64 | 20.64 | 74.17 | 18.32 | 98.25 |
| Feb | 3.12 | 1.34 | 31.99 | 1.29 | 55.76 | 7.73 | 75.45 | 17.16 | 81.91 |
| Mar | 6.52 | 18.28 | 38.64 | 30.44 | 45.86 | 16.55 | 86.96 | 39.53 | 85.13 |
| Apr | 12.84 | 4.06 | 16.03 | 0.02 | 20.59 | 7.36 | 52.39 | −14.53 | 87.63 |
| May | 19.00 | 11.25 | 26.20 | 22.09 | 45.86 | 18.12 | 39.65 | 17.95 | 49.72 |
| Jun | 20.29 | 0.76 | 17.40 | 2.69 | 27.51 | 14.39 | 11.93 | 3.81 | 17.07 |
| Jul | 20.09 | 10.72 | 30.57 | 7.74 | 25.48 | 0.77 | −5.17 | 7.16 | 21.31 |
| Aug | 16.10 | −9.44 | 15.22 | −8.74 | 1.59 | 5.33 | −19.02 | −6.34 | −5.63 |
| Sep | 13.77 | −18.72 | −5.28 | −20.14 | −1.06 | −20.96 | 25.68 | −5.95 | −27.96 |
| Oct | 9.13 | 16.10 | −0.34 | 21.60 | 11.07 | 23.57 | 5.51 | 5.65 | 31.98 |
| Nov | 5.10 | 21.11 | 34.81 | 40.76 | 48.80 | 40.63 | 69.85 | 37.81 | 72.51 |
| Dec | 4.11 | 19.64 | 9.59 | 18.87 | 40.77 | 7.78 | 69.85 | 12.15 | 79.55 |
| Annual | 133.15 | 4.00 | 18.55 | 7.09 | 24.91 | 8.96 | 19.74 | 6.41 | 33.56 |

Note:"MC" represents the mid-21st century (2050 s), and "LC" denotes the late-21st century (2100 s).

trend is expected to become more pronounced in the future as carbon emissions increase. Please see the appendix for the results on future climate variables that were not analysed (Appendix Table 1-2 and Appendix Fig. 4-9).

Please see the appendix for results on future climate variables, which were not analyzsd for length reasons (Appendix Table 1 and 2 and Appendix Fig. 4- 9).

### 3.3. VIC model simulation assessment

To achieve a full modelling spin-up, the simulation period was initiated in 1959. The model was calibrated and validated using simulations from 1961 to 2014, with a calibration period spanning 1961 to 1985 and a validation period from 1985 to 2014. The VIC model was successfully calibrated to reproduce the runoff and evapotranspiration observed during the baseline period (1961–2014). The optimised performance parameters and hydrologic scatter plots are presented in Table 2, Figs. 6 and 7. Further details are provided for the Bole and Wenquan hydrologic stations in Tables 3 and 4, respectively, in the Appendix.

Compared with the HGCM data before downscaling, the performance of the BC data-driven model yielded satisfactory results for NSE, RMSE, BIAS, $R^2$, and R (Table 2 and Appendix Table 3 and 4; Fig. 6g–i). At Jinghe Hydrological Station, calibration sets achieved values of 0.81 (0.89), 6.14 mm (4.78 mm), −0.09 mm (−0.08 mm), 0.83 (0.90), and 0.91 (0.95). Meanwhile, the validation sets yielded values of 0.82 (0.89), 5.81 mm (4.51 mm), 0.02 mm (0.01 mm), 0.84 (0.90), and 0.92 (0.95) with performance parameters for the TPS station shown in parentheses (Table 2 and Fig. 6g). Better simulation results were obtained for the remaining two hydrographic stations. The accuracy of the runoff simulation improved, on average, by approximately 130 % during the calibration period and by approximately 128 % during the validation period compared with the pre-downscaling conditions (Appendix Table 3 and 4 and Fig. 6h–i). Satisfactory results were also obtained for the simulated performance parameters of evapotranspiration, with $R^2$, NSE, and RMSE values of 0.87 (0.53), 0.86 (0.50), and 48.40 mm (90.43 mm), respectively. The performance parameters for the TPS station are shown in parentheses (Fig. 7c). On average, the accuracy of the evapotranspiration simulation improved by approximately 86.79 % compared to the pre-downscaled conditions. A comparison of the inter- and intra-annual hydrographs showed that the simulated runoff and evapotranspiration after downscaling showed stronger alignment with the observed values for both low- and high-frequency fluctuations (Fig. 6a–c) and Fig. 7a). Larger runoff fluctuations were observed at the Jinghe Hydrological Station, whereas a distinctive intra-annual runoff distribution was noted at the Bole hydrological station (Fig. 6d–f). The CDF distribution plots demonstrate that the simulated values after QDM correction achieved a well-fitted distribution within the middle and low numerical intervals (Fig. 6j–l) and Fig. 7d).

The QDM-corrected driving simulation values were close to the TPS station driving simulation values. Both effectively reproduced the observed runoff and evapotranspiration trends during the historical period. This established a reliable foundation for simulating hydrological fluxes and estimating future eco-agricultural water consumption.

**Table 2**
Performance evaluation of runoff simulation at Jinghe hydrological station (Monthly Scale).

| Input Data Type | NSE | RMSE | BIAS | $R^2$ | R |
|---|---|---|---|---|---|
| TPS_station(Calibration) | 0.89 | 4.78 | −0.08 | 0.90 | 0.95 |
| TPS_station(Validation) | 0.89 | 4.51 | 0.01 | 0.90 | 0.95 |
| BC data (Calibration) | 0.81 | 6.14 | −0.09 | 0.83 | 0.91 |
| BC data (Validation) | 0.82 | 5.81 | 0.02 | 0.84 | 0.92 |
| HGCM (Calibration) | 0.03 | 14.06 | 0.25 | 0.03 | 0.18 |
| HGCM (Validation) | 0.06 | 13.27 | −0.29 | 0.07 | 0.26 |

Note:TPS_station, BC data, and HGCM represent interpolated meteorological station data, QDM bias-corrected data, and raw GCM output data, respectively.

### 3.4. Changes in agricultural and ecological water consumption under future scenarios

After the water consumption calculation programme was executed, simulated agricultural and ecological water consumption in the study area was obtained. The grid-scale water consumption (mm) was then converted to the total water volume ($m^3$) and analysed in terms of relative change differences and spatiotemporal change trends.

#### 3.4.1. Relative change in water consumption

Relative to the baseline period, the simulated agricultural water consumption, that is, the ecological water consumption changed by 8.00 % (13.31 %), 10.88 % (15.03 %), 11.66 % (16.33 %), and 9.02 % (15.86 %) in the mid-21st century under the SSP126 to SSP585 scenarios. In the late 21st century, changes of 21.43 % (25.18 %), 29.78 % (34.34 %), 29.91 % (35.48 %), and 43.13 % (49.11 %) were observed (Table 3–4). In terms of monthly changes, agricultural water consumption exhibited a relative uptrend for most of the MC period compared with the baseline. The most pronounced increase was observed from March to May, reaching up to 24.08 % in April under the SSP370 scenario. During the LC period, this relatively positive change became even more pronounced. It largely corresponded to the rise in carbon emission scenarios, peaking at 101.29 % in April under the SSP585 scenario. Ecological water consumption showed a decreasing

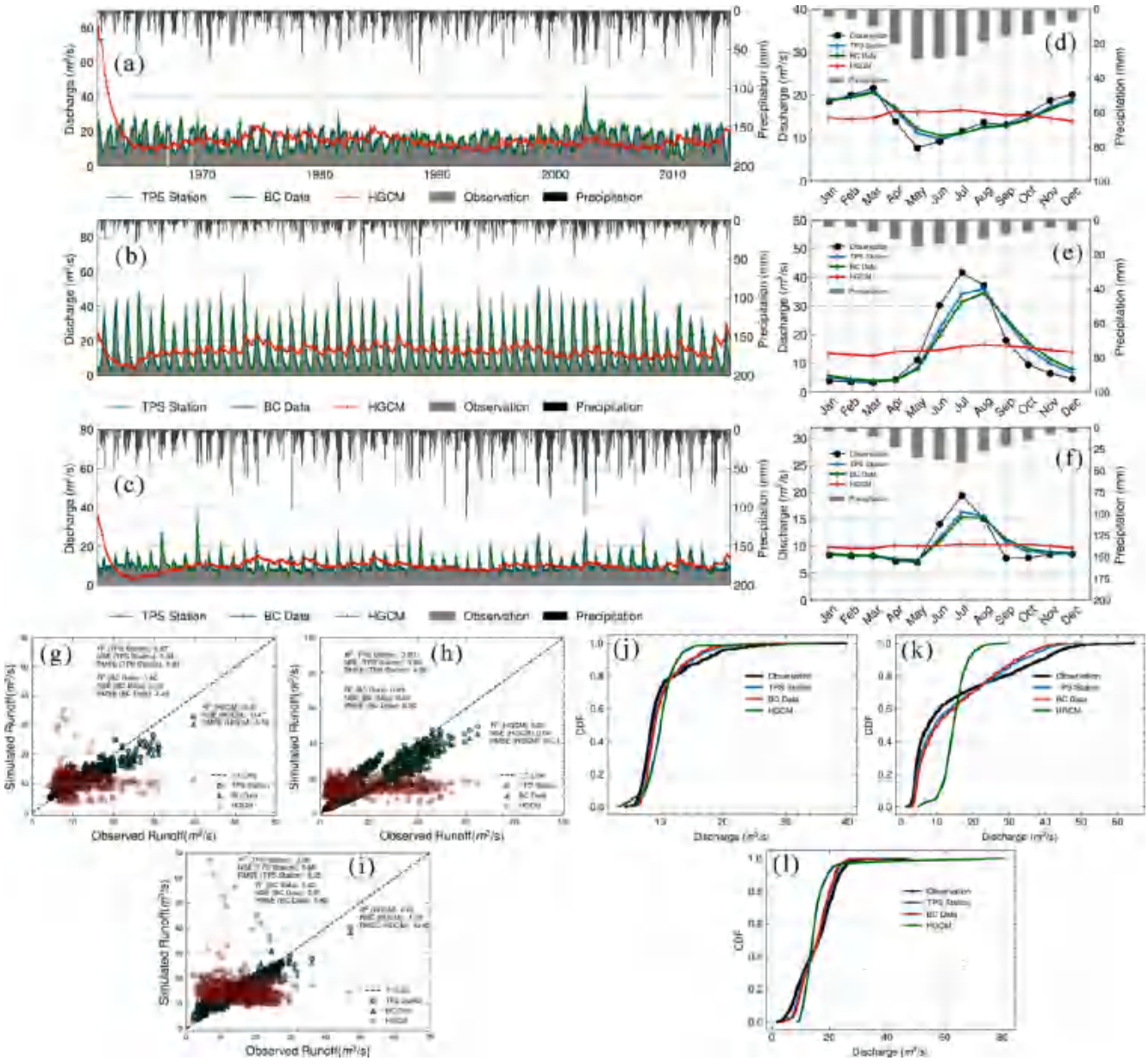


**Fig. 6.** Comparison of simulated and observed runoff through hydrographs, scatter Plots, and CDF distributions. (Panels (a) to (c) display the inter-annual hydrographs for the Bole, Jinghe, and Wenquan hydrological stations, while panels (d) to (f) illustrate the intra-annual hydrographs for the same stations. Panels (g) to (i) present scatter plots comparing simulated and observed runoff at Bole, Jinghe, and Wenquan stations, respectively. Panels (j) to (l) show the Empirical Cumulative Distribution Function (CDF) distributions of simulated and observed runoff. "Observation" refers to observed runoff; "TPS Station" represents runoff simulated by the VIC model using TPS interpolated meteorological data; "BC Data" indicates runoff simulated using QDM bias-corrected climate model data; and "HGCM" corresponds to runoff simulated using the original climate model data.).

trend during December, January, and February compared to the baseline period, with declines of − 20.48 % and − 11.72 % observed in January under the SSP585 scenario during the MC and LC periods, respectively. An upward trend was observed in most of the remaining months with increases of 10.54 % and 32.86 % recorded in July under the SSP585 scenario during the MC and LC periods, respectively (Table 4). In summary, agricultural water consumption increased with the intensification of human activities, while ecological water consumption decreased in December, January, and February but increased throughout the year, remaining higher than agricultural water consumption.

### 3.4.2. Change in total water consumption over time

To provide a clearer comparison of future scenarios for agricultural and ecological water consumption, the average annual water consumption (billion $m^3$) was calculated for both categories. Under the mid-21st century baseline and scenarios SSP126 to SSP585, agricultural water consumption was 0.34, 0.37, 0.38, 0.38, and 0.37 billion $m^3$, respectively. Meanwhile, ecological water consumption was significantly higher, at 4.40, 4.98, 5.06, 5.11, and 5.09 billion $m^3$, respectively (Fig. 8a–b). For late 21st-century scenarios, agricultural water consumption increased to 0.41, 0.44, 0.44, and 0.49 billion $m^3$ under SSP126, SSP245, SSP370, and SSP585, respectively. Ecological water consumption increased to 5.50, 5.91, 5.96, and 6.56 billion $m^3$, respectively (Fig. 8a–b). Agricultural and ecological water consumption increased across all SSP scenarios, with agricultural water consumption rising by 8.82–11.76 % in the mid-21st century and by 20.59–44.12 % in the late 21st century, relative to the baseline period. Similarly,

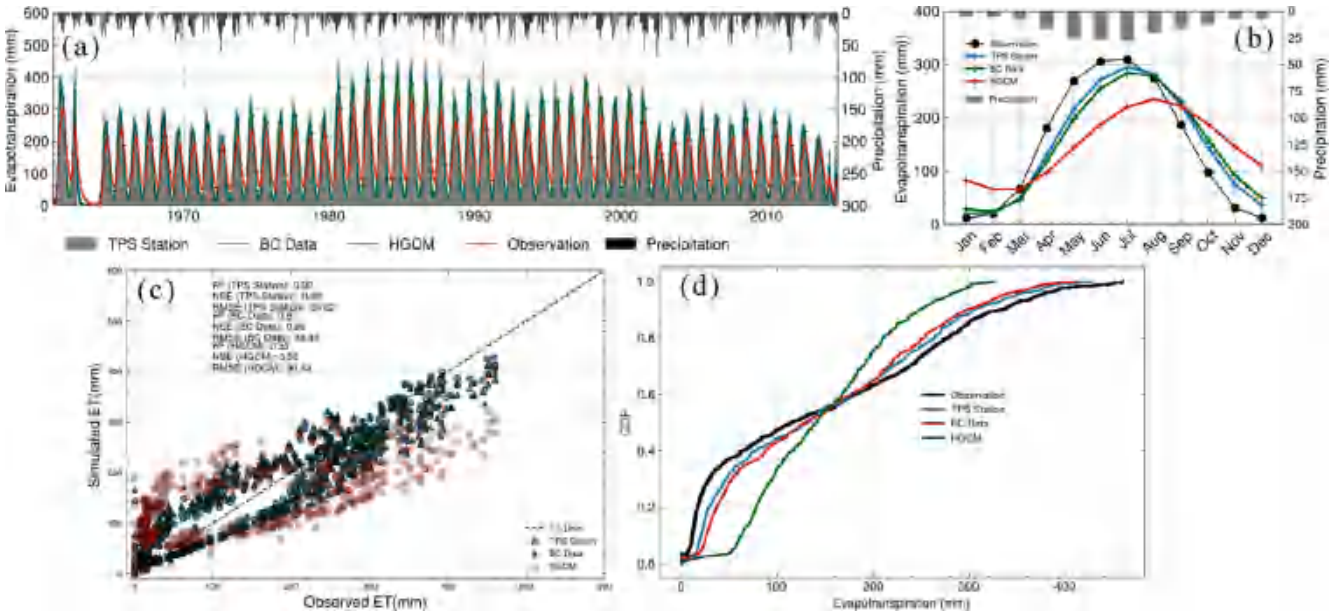


**Fig. 7.** Comparison of simulated and observed evapotranspiration through hydrographs, scatter plots, and CDF distributions. (Panel (a) displays the inter-annual time series hydrographs of average evapotranspiration values, while panel (b) shows the intra-annual time series hydrographs for the same averages. Panel (c) presents scatter plots comparing simulated and observed evapotranspiration for average values. Panel (d) illustrates the Empirical Cumulative Distribution Function (CDF) distributions for the comparison between simulated and observed evapotranspiration. "Observation" refers to observed evapotranspiration; "TPS Station" indicates evapotranspiration simulated by the VIC model driven by TPS interpolated meteorological data; "BC Data" denotes evapotranspiration simulated using QDM bias-corrected climate model data; and "HGCM" represents evapotranspiration simulated using the original climate model data.).

**Table 3**
Projected relative changes in agricultural water consumption by the mid-21st century (2050 s) and late 21st century (2100 s) under different SSP scenarios, relative to the baseline period (1961–2014).

| Month | Baseline | Relative change in Agricultural Water Consumption (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | (Million $m^3$) | SSP126 | | SSP245 | | SSP370 | | SSP585 | |
| | | MC | LC | MC | LC | MC | LC | MC | LC |
| Jan | 4.53 | 2.39 | 0.03 | 10.63 | −3.33 | 10.51 | 14.55 | 1.13 | 19.45 |
| Feb | 9.66 | 0.01 | 1.16 | 10.04 | −7.37 | 5.34 | −4.25 | 1.52 | −4.35 |
| Mar | 15.72 | 3.63 | 17.55 | 13.01 | 21.66 | 6.59 | 49.07 | 17.34 | 64.99 |
| Apr | 30.66 | 19.91 | 37.42 | 21.77 | 47.96 | 24.08 | 80.06 | 13.47 | 101.29 |
| May | 52.28 | 19.66 | 34.58 | 23.71 | 48.25 | 18.72 | 54.62 | 22.48 | 61.11 |
| Jun | 52.28 | 4.79 | 13.08 | 5.63 | 28.11 | 8.99 | 21.95 | 0.59 | 33.70 |
| Jul | 64.33 | 3.39 | 17.92 | 6.98 | 25.41 | 2.69 | 10.85 | 2.91 | 25.61 |
| Aug | 45.55 | 3.23 | 21.58 | 4.33 | 22.07 | 15.58 | 13.62 | 7.92 | 31.06 |
| Sep | 25.54 | 5.69 | 28.65 | 5.39 | 37.55 | 11.80 | 24.32 | 16.02 | 39.77 |
| Oct | 12.66 | 10.74 | 10.71 | 10.16 | 21.88 | 14.81 | 34.67 | 6.15 | 50.02 |
| Nov | 5.95 | −1.13 | 11.65 | 5.92 | 11.24 | 7.06 | 16.99 | 12.65 | 34.02 |
| Dec | 5.47 | 9.41 | 15.94 | 21.12 | 6.21 | 4.51 | 18.70 | 7.01 | 19.12 |
| Annual | 340.15 | 8.00 | 21.43 | 10.88 | 29.78 | 11.66 | 29.91 | 9.02 | 43.13 |

ecological water consumption increased by 13.18–16.14 % in the mid-21st century and by 25.00–49.09 % in the late 21st century, with the largest increase observed under the SSP585 scenario. These trends illustrate a consistent increase in water demand in future scenarios, particularly in the context of higher carbon emissions. From the intra-annual distribution, both agricultural and ecological water consumption aligned with the typical water use patterns observed in arid regions, with peak water consumption occurring between May and August (Fig. 8c–f). Under the mid-21st century scenarios, agricultural water consumption peaked in June at 0.076 billion $m^3$ under SSP370. Meanwhile, ecological water consumption reached its maximum in June at 110 billion $m^3$ under SSP245 (Fig. 8c–d). For the late 21st century, agricultural water consumption peaked in June at 0.086 billion $m^3$ under SSP585, whereas ecological water consumption reached its highest level in July at 1.25 billion $m^3$ under SSP585 (Fig. 8e–f). These results highlight seasonal and scenario-specific variations in water consumption patterns.

### 3.4.3. *Temporal and spatial change in grid-scale water consumption*

Agricultural water consumption was relatively high across the plains surrounding Lake Ebinur and along both sides of the Jinghe and Bortala Rivers. Meanwhile, the rest of the area exhibited relatively low values (Fig. 9A–B). This spatial distribution pattern showed no significant changes in future projections. In terms of absolute spatial changes, positive changes were identified across most of the region under all scenarios in the mid- and late 21st century compared with the baseline period. Only minor negative changes were observed east of Tailrace Lake (Fig. 9C). The mean absolute change was higher in the late 21st century than in the mid-21st century, with the maximum value in the mid-21st century recorded at 2.11 mm under SSP370 and the maximum in the late 21st century reaching 7.67 mm under SSP585 (Fig. 9C). Most of this positive change was concentrated in the cultivated areas of the

**Table 4**
Projected relative changes in ecological water consumption by the mid-21st century (2050 s) and late 21st century (2100 s) under different SSP scenarios, relative to the baseline period (1961–2014).

| Month | Baseline | Relative change in Ecological Water Consumption (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | (Million $m^3$) | SSP126 | | SSP245 | | SSP370 | | SSP585 | |
| | | MC | LC | MC | LC | MC | LC | MC | LC |
| Jan | 39.77 | −14.09 | −15.62 | −12.41 | −19.42 | −7.98 | −3.19 | −20.48 | 11.72 |
| Feb | 94.96 | −1.80 | −1.90 | 4.59 | −6.59 | 2.10 | −0.98 | −3.42 | −1.50 |
| Mar | 174.79 | 5.78 | 8.96 | 8.60 | 7.14 | 6.25 | 25.43 | 13.76 | 36.41 |
| Apr | 293.48 | 21.33 | 34.03 | 18.23 | 49.60 | 26.41 | 85.03 | 17.90 | 111.90 |
| May | 572.58 | 26.49 | 43.52 | 31.68 | 59.68 | 28.14 | 64.87 | 35.65 | 76.87 |
| Jun | 891.24 | 12.90 | 20.83 | 13.84 | 33.97 | 17.71 | 32.55 | 12.09 | 43.30 |
| Jul | 983.38 | 10.55 | 22.87 | 12.86 | 28.86 | 9.80 | 21.38 | 10.54 | 32.86 |
| Aug | 745.94 | 9.68 | 24.84 | 10.51 | 29.78 | 16.50 | 22.11 | 13.76 | 38.18 |
| Sep | 347.17 | 15.01 | 33.79 | 16.32 | 48.38 | 17.30 | 38.03 | 25.05 | 57.15 |
| Oct | 146.77 | 16.22 | 15.05 | 9.61 | 24.49 | 16.24 | 46.96 | 10.95 | 62.57 |
| Nov | 65.53 | 2.24 | 15.70 | 14.39 | 14.85 | 10.98 | 28.26 | 17.59 | 39.07 |
| Dec | 40.90 | −0.41 | 11.34 | 6.81 | 4.41 | 3.10 | 29.33 | −1.24 | 34.43 |
| Annual | 4396.53 | 13.31 | 25.18 | 15.03 | 34.34 | 16.33 | 35.48 | 15.86 | 49.11 |

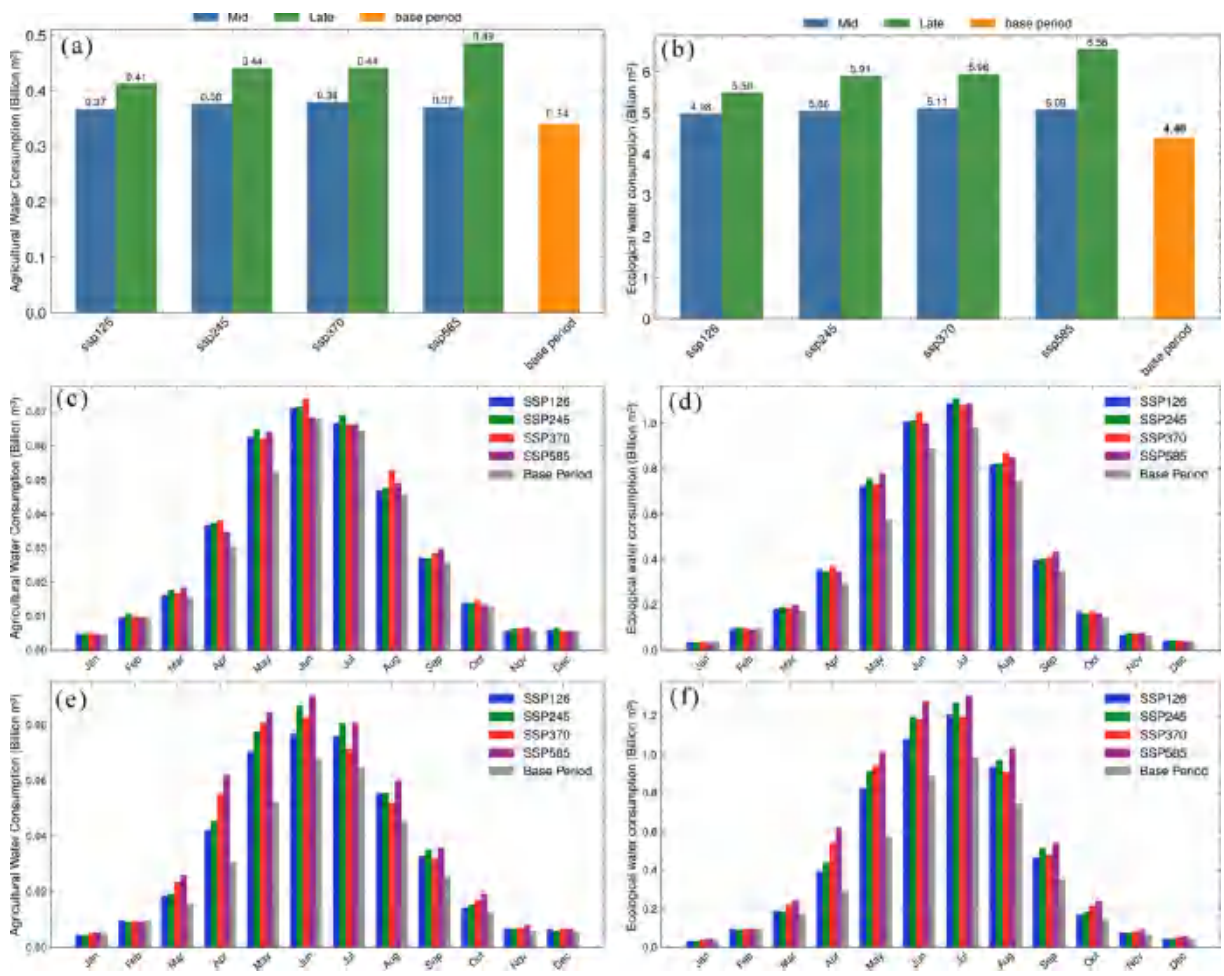


**Fig. 8.** Temporal and statistical distributions of simulated agricultural and ecological water consumption across different future SSP scenarios. ((a) shows the inter-annual distribution of agricultural water consumption, while (b) presents the inter-annual distribution of ecological water consumption across various SSP scenarios. (c) and (d) display the intra-annual distributions of agricultural and ecological water consumption for the near-term period, respectively. (e) and (f) depict the intra-annual distributions of agricultural and ecological water consumption for the long-term period.).

plains along the riverbanks (Fig. 9C). Ecological water consumption exhibited a higher spatial distribution in the near-mountain plains and a lower distribution along the riverbanks and central plains, with these spatial patterns remaining unchanged across all future scenarios (Fig. 10A–B). This pattern can be attributed to the high ecological water consumption driven by the presence of deciduous, coniferous forests and meadows in near-mountain areas. Ecological water consumption in the mid-21st century showed a lower absolute change compared with the baseline period than in the late 21st century. Most areas exhibited positive changes concentrated in the mountainous and near-mountain plains of the basin (Fig. 10C). The mean absolute change reached a maximum of 20.57 mm in the mid-21st century under SSP585 and

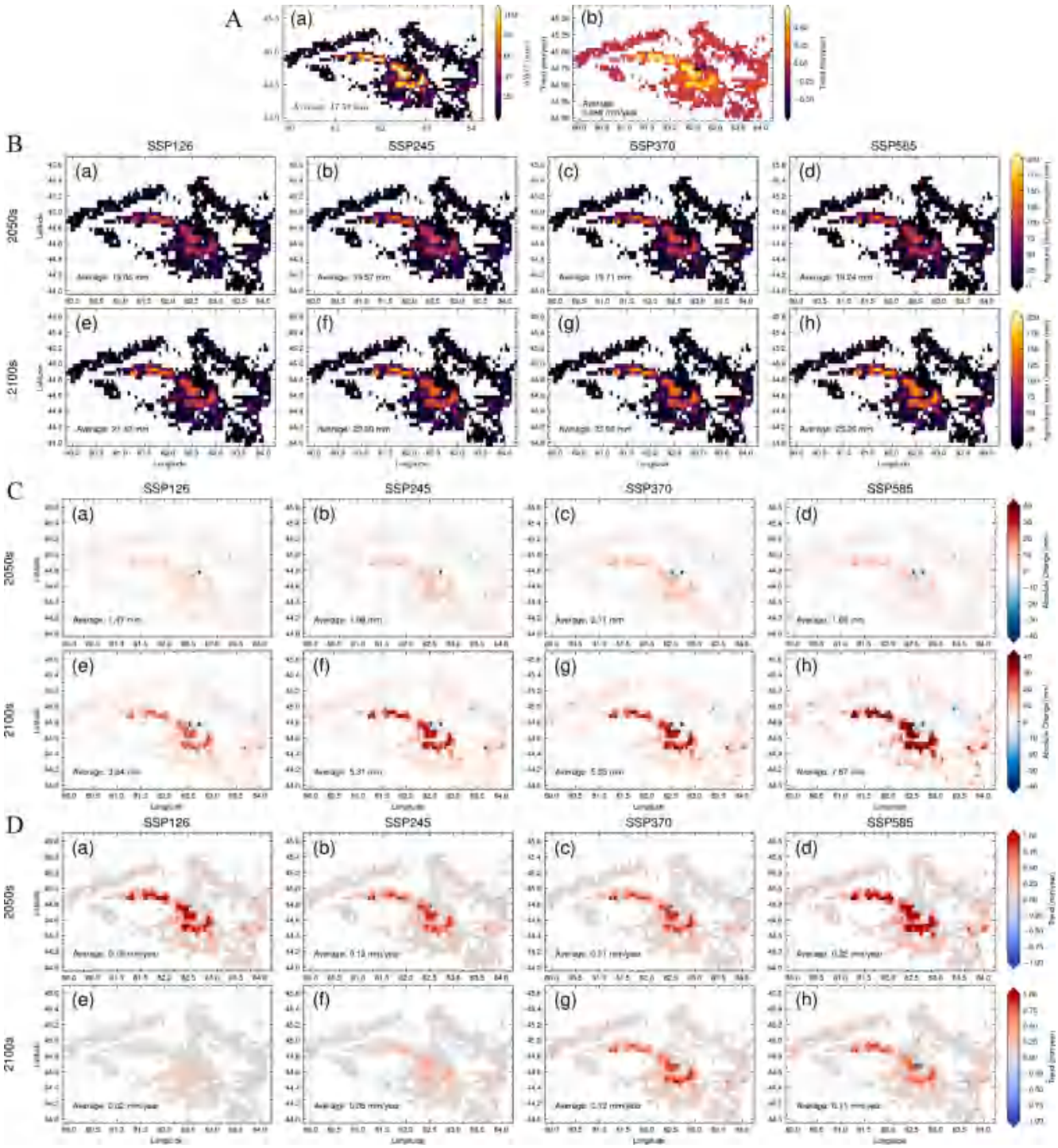


**Fig. 9.** Annual average spatial distribution and trends of simulated agricultural water consumption under different SSP scenarios in the near and distant future, including absolute spatial differences compared to the baseline. (Panel A-a shows the annual average spatial distribution of agricultural water consumption during the base period, while A-b illustrates the corresponding spatial trend. Panel B presents the annual average spatial distribution under different SSP scenarios. Panel C depicts the absolute spatial differences in annual average agricultural water consumption under various SSP scenarios compared to the base period. Panel D displays the spatial trends of agricultural water consumption under different SSP scenarios.).

62.36 mm in the late 21st century under the same scenario (Fig. 10C). The rate of spatial trend change indicated that agricultural water consumption could reach a maximum of 1.24 mm/year in the cultivated areas of the Central Plains during the mid-21st century and 1.05 mm/year during the late 21st century. The mean consumption rates were recorded at 0.22 mm/year in the mid-21st century under SSP585 and 0.12 mm/year in the late 21st century under SSP370 (Fig. 9D). Ecological water consumption reached maximum rates exceeding 2.00 mm/year in the mountain forest region, with mean rates peaking at 1.32 mm/year in the mid-21st century and 0.82 mm/year in the late 21st century under the SSP585 scenario (Fig. 10D). These findings suggest that the rate of increase in the late 21st century was slower than in the mid-21st century.

The results of intra-annual temporal changes in agricultural water

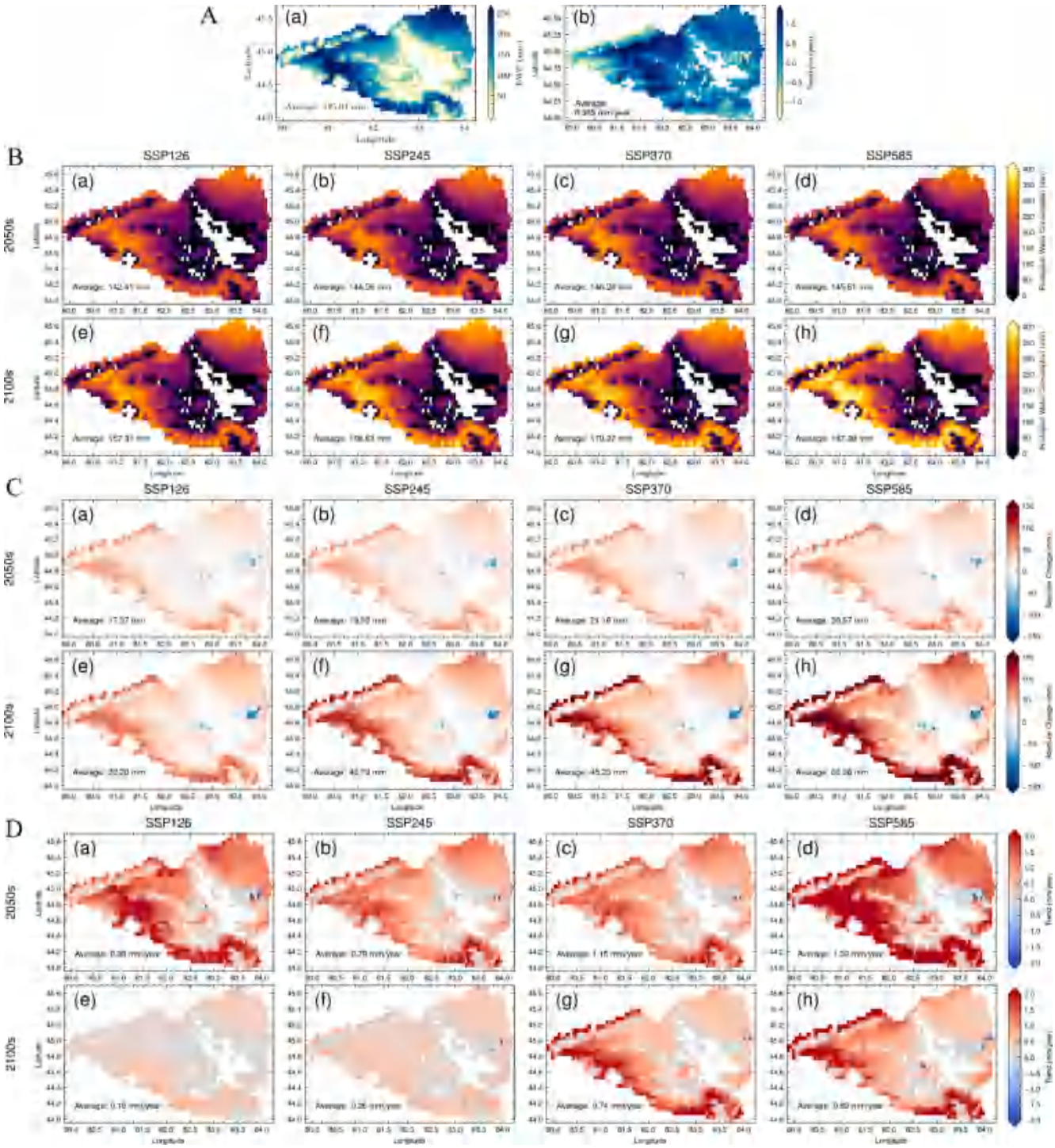


**Fig. 10.** Annual average spatial distribution and trends of simulated ecological water consumption under different SSP scenarios in the near and distant future, including absolute spatial differences compared to the baseline. (Panel A-a shows the annual average spatial distribution of ecological water consumption during the base period, while A-b illustrates the corresponding spatial trend. Panel B presents the annual average spatial distribution under different SSP scenarios. Panel C depicts the absolute spatial differences in annual average ecological water consumption under various SSP scenarios compared to the base period. Panel D displays the spatial trends of ecological water consumption under different SSP scenarios.).

consumption indicate that SSP585 exhibits the highest distribution in the late season, with consumption approximately 42 % higher than the baseline period during summer, followed by the other scenarios (Fig. 11A-a). The inter-annual time-varying results indicate that agricultural water consumption exhibits a gradual upward trend, with a relatively faster increase observed under the SSP585 scenario after 2050 (Fig. 11A-b and A-c). The corresponding boxplot indicates that the highest mean value was observed under SSP585 at approximately 10.58 mm, accompanied by the widest range. Meanwhile, the lowest mean value was recorded under SSP126 at approximately 9 mm, with the narrowest range (Fig. 11A-d). The temporal variation in the mean distance value indicated that it remained above 0 mm after 2014, gradually increasing to a range of 2–4 mm, with the SSP585 scenario showing the highest values (Fig. 11A-e and A-f). The corresponding boxplot showed

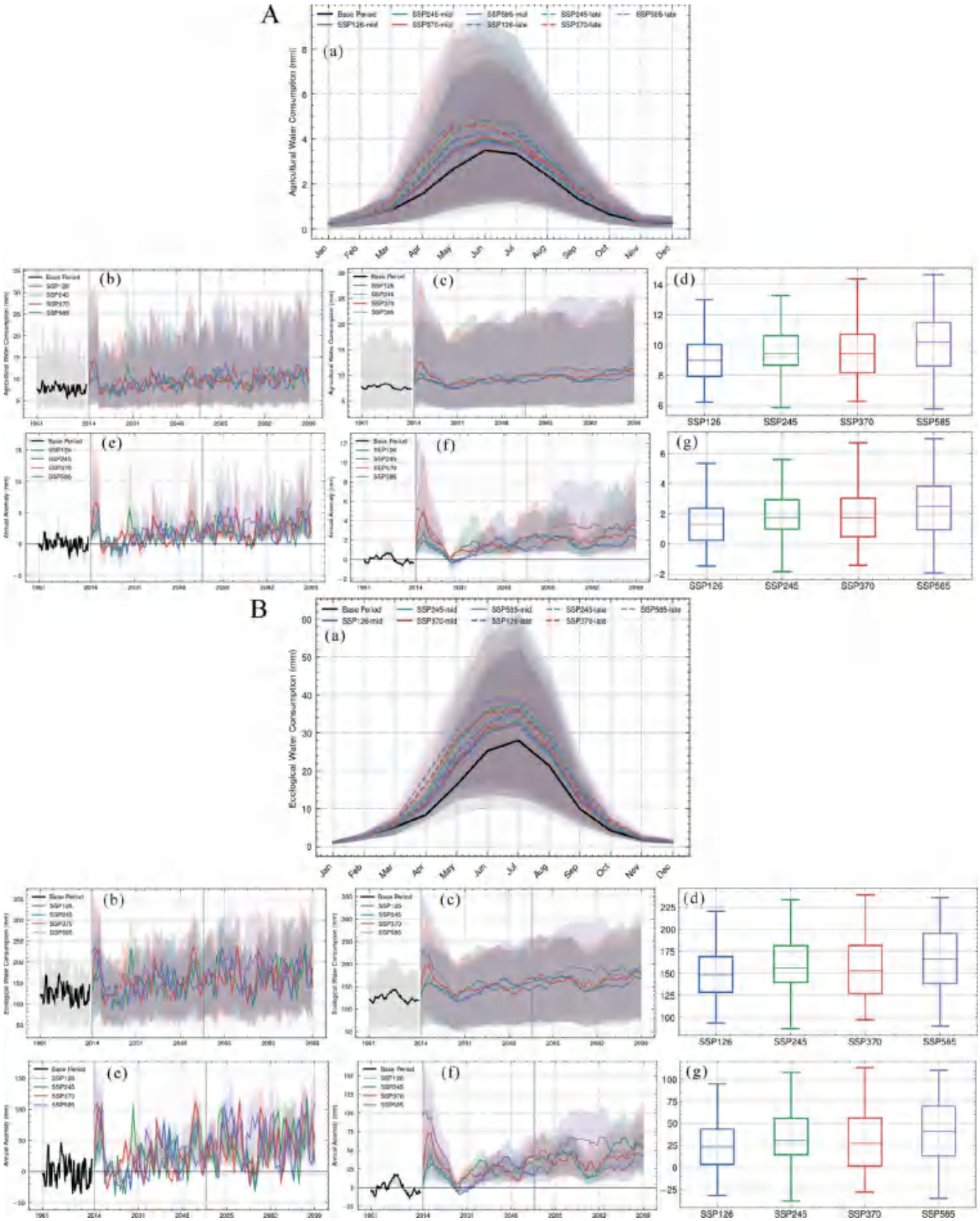


**Fig. 11.** Time-Series Trends of Differences in Annual Average Agricultural and Ecological Water Consumption Relative to the Baseline Period Under Different SSP Scenarios. (A and B represent simulated intra- and inter-annual trends of agricultural and ecological water consumption, respectively, along with their corresponding box plots. The intra-annual trends (a) show seasonal variations, while inter-annual trends (b–d) display the changes in mean annual water consumption, 10-year moving averages, and boxplot distributions compared to the baseline. Panels (e–g) further illustrate inter-annual differences, highlighting long-term trends and variability in mean annual water consumption under varying SSP scenarios.).

that the mean value for SSP585 was the highest at 2.5 mm, accompanied by the widest range. Meanwhile, SSP126 had the lowest mean value at 1.7 mm and the narrowest range (Fig. 11A-g). The intra-annual temporal variation in ecological water consumption exhibited a similar pattern, with the SSP585 scenario reaching a maximum value of 39.80 mm in July during summer, representing an average increase of 40 % compared with the baseline period. The other scenarios show values above those of the baseline period (Fig. 11B-a). Interannual temporal variations indicated that ecological water consumption under the SSP585 scenario was projected to be relatively high after 2050, stabilising within a range of approximately 150–200 mm. Meanwhile, the other scenarios showed a slower and more gradual increase (Fig. 11B-b and B-c). The corresponding boxplot showed that the highest mean value was observed under the SSP585 scenario at approximately 170 mm, accompanied by a wider range. Meanwhile, the lowest mean value was recorded under SSP126 at approximately 148 mm, with a narrower range (Fig. 11B-d). The temporal changes in the mean value of the distance showed a significant drop from 100 mm to 0 mm between 2014 and 2031, followed by a gradual increase to the 50–75 mm range. SSP585 consistently remained higher than the other scenarios for most future periods (Fig. 11B-e and B-f). Agricultural and ecological water consumption will continue to rise in the future, with the rate of increase being primarily driven by intensified human activities.

The results indicated that agricultural and ecological water consumption were projected to continue rising in the future, with the rate of increase primarily driven by the trend of intensified human activities.

## 4. Discussion

### 4.1. Uncertainty analysis

In this study, changes in agricultural and ecological water consumption across various SSP scenarios were investigated, focusing on the baseline period and future projections. These investigations were primarily conducted using simulations based on the VIC model. However, some inherent uncertainties are associated with the simulated runoff and evapotranspiration calculations, downscaling of CMIP6 data, and methods used for estimating water consumption. The largest source of uncertainty in predicting and analysing future hydrological climate change originates from General Circulation Models. This uncertainty is further compounded by factors such as downscaling algorithms and hydrological models. The uncertainties associated with the hydrological model used in this study could be attributed to several factors. Firstly, the estimation of agricultural water consumption was subject to uncertainty because of the simplified structure of the irrigation module. This failed to adequately capture the heterogeneity of irrigation management practices at the basin and regional scales. Secondly, the absence of real-time updated land-cover data assimilation into the hydrological model further contributes to uncertainty in future hydrological simulations. Due to time and resource constraints, future efforts will focus on refining the model structure to develop a hydrological model specifically tailored to inland arid regions. An uncertainty analysis was performed using the Differential Evolution Adaptive Metropolis (DREAM) algorithm, combined with a parameter sensitivity analysis conducted using the Fourier Amplitude Sensitivity Test (FAST) method, enabling the estimation of the approximate range of simulation uncertainty and the identification of sensitive parameters. The differential evolution adaptive metropolis (DREAM) algorithm integrates the differential evolution and adaptive metropolis-Hastings approaches, enabling improved efficiency in exploring high-dimensional parameter spaces. In contrast, the FAST methodology was used for sensitivity analysis. The resulting posterior simulation uncertainty and parameter sensitivity outcomes for the VIC model are illustrated in Fig. 12.

The analysis of the a posteriori simulation uncertainty range showed that for the Bole hydrological station, the runoff simulation uncertainty was approximately 30 % above and below the measured value (Fig. 12a). At the Jinghe station, the uncertainty was relatively small, approximately 10 %, while a similar uncertainty of approximately 30 % was observed at the Wenquan station (Fig. 12b–c). The results of the parameter sensitivity analysis revealed distinct patterns across the different hydrological stations, with the parameters Ds and Soil_D3 identified as more sensitive at the Bole hydrological station (Fig. 12d). In contrast, at the Jinghe station, the parameters binfill, Ws, and Soil_D3 exhibited higher sensitivity. Meanwhile, at the Wenquan station, binfill, C, and Soil_D1 were identified as more sensitive. These findings align with those reported by Xie et al. (2007) (Fig. 12e–f). This consistency highlights the suitability of the approach used in this study for future large-scale research in arid regions.

Although the CMIP6 model output suitable for the inland arid regions was selected in this study based on the findings of Jiang et al. (2020), the predicted range of future changes in precipitation, air temperature, and wind speed remained significantly large compared to the baseline period. Compared with the baseline period, projected temperature increases were estimated at 67.02 % (94.69 %), 75.02 % (136.83 %), 80.30 % (193.99 %), and 84.87 % (230.88 %) under SSP126 to SSP585 for the mid- and late-21st century, respectively. The ranges of simulated changes in future runoff and evapotranspiration are expected to be correspondingly larger, driven by a wider range of relative changes

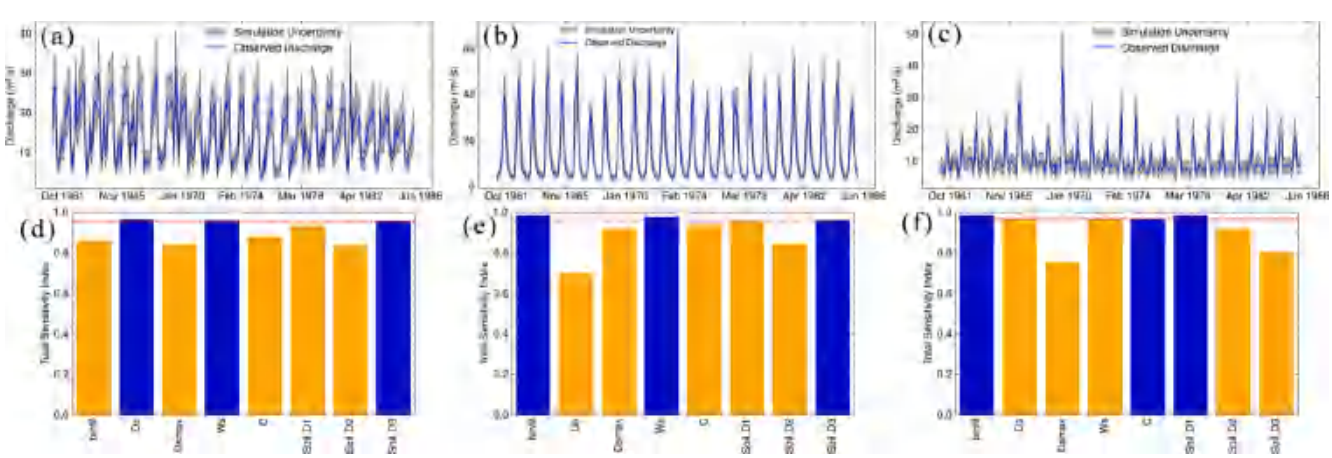


**Fig. 12.** A posteriori uncertainty analysis of the VIC hydrological model, with sensitivity results obtained using the FAST method to evaluate the impact of key input parameters on model performance. (Panels a to c represent the simulated uncertainty ranges for Bole, Jinghe, and Wenquan hydrological stations, respectively. Panels d to f show the total sensitivity indices of the parameters for the same stations, where sensitive parameters are highlighted in blue and insensitive parameters in orange.).

in meteorological variables. Uncertainty in hydrologic predictions is also influenced by the choice of the downscaling algorithms, with the QDM statistical downscaling method widely applied in runoff simulation and prediction. Kim et al. (2021) demonstrated that the QDM approach, which accounts for differences in the time-dependent (autocorrelation) structure between downscaled GCMs and observed rainfall, is most effective in reducing biases introduced into sequencing statistics. The results of the data-driven simulations using the QDM bias correction were closely aligned with the interpolated results from the TPS meteorological sites, supporting the validity of the historical and future projected series of meteorological variables after downscaling.

While machine learning models, WEAP-MODFLOW, and CMIP6 climate outputs were used by Jamnani et al. (2024) to estimate evapotranspiration and groundwater changes, separate evaluations of agricultural and ecological water consumption were not performed. In this study, the application of QDM bias correction improved the accuracy of CMIP6 historical meteorological data by approximately 71.66 %. The performance of the VIC model simulations for runoff and evapotranspiration were enhanced by approximately 80.43 % and 63.61 %, respectively. Kim et al. (2021) applied the regional QDM method for bias correction of GCM output data, reporting an average error ranging from − 2.11 % to 0.70 % compared with observational data. Robertson et al. (2023) compared four quantile mapping methods and found that the QDM approach reduced runoff simulation bias by 2 %. Despite the advantages of the downscaling algorithm, although Zhai et al. (2022) estimated changes in agricultural, industrial, domestic, and ecological water use under future warming scenarios, the use of different models increased the complexity of the parameter calibration and optimisation processes. By contrast, the advantage of this study lies in the use of a VIC-based simulation separation framework that streamlines the calibration and validation process and achieves superior accuracy. The method used in this study was differentiated by the integration of a VIC-based simulation separation framework with QDM bias correction. This allowed for precise and efficient assessments of agricultural and ecological water consumption, while ensuring high accuracy and optimising the calibration and validation processes.

### 4.2. Reflections on the deficits between water supply and water consumption

Although extensive research has been conducted on the impacts of future climate change on surface water resources, including river and lake runoff, there have been relatively few studies on the dynamics of water supply and usage balance in the basins of arid inland regions. There has also been relatively little research on the combined influence of climate change and agricultural development in this context. There is a notable scarcity of studies that integrate physical process models with GCMs to examine the disparities in water supply and water balance in inland arid regions. Water supply, withdrawal, and security in China's basins were examined by Zhai et al. (2022) using the VIC model, validated with measured runoff. However, their study did not incorporate changes in agricultural and ecological water consumption, relying instead on statistical yearbook data that omitted a detailed explanation of the physical process.

Located in an arid region of northwestern China, the Lake Ebinur basin is a critical example of the challenges faced in balancing water resource allocation between agricultural demand and ecological conservation. However, the region has been facing severe land degradation and environmental challenges due to water scarcity. There has been an increasing trend of land degradation over the past two decades, primarily driven by human activities, alongside a significant decline in water yield between 2000 and 2020 (Yang et al., 2024). These findings highlight the research value of this study area, where water scarcity has been exacerbated by human activities and is primarily driven by agricultural demand. The ecological degradation and land deterioration caused by severe water shortages are expected to intensify in the future, posing substantial risks to regional sustainability and agricultural productivity.

The restoration of cropland method to achieve a balance between the restoration of Lake Ebinur and human water use was proposed by Zeng et al. (2019), who demonstrated that restoring the lake's surface area to 552 km2 (800 km2) would require an increase in annual inflow to 602.6 $\times 10^6$ m3 (876.6 $\times 10^6$ m3) and a reduction in human water use by 320 $\times 10^6$ m3 (594.2 $\times 10^6$ m3). Although the restoration goals for Lake Ebinur were highlighted by Zeng et al. (2019), the water balance gap in the basin was not addressed nor were physical processes incorporated. In contrast, changes in total water resources and agricultural and ecological water consumption were estimated in our study for Lake Ebinur basin. An innovative framework for calculating water consumption in agriculture and natural vegetation has been proposed based on a physical process model that uses the VIC model in conjunction with the latest CMIP6 GCM outputs. The model was calibrated and validated using measured data from meteorological and hydrological stations. The total water resources and socioeconomic water consumption for different future scenarios were estimated primarily using historical data from local statistical yearbooks. These estimates are based on historical water availability (40 %) and the ratio of agricultural water use to socioeconomic water consumption (80 %) (Han and Jia, 2022). The deficit between water supply and consumption under various SSPs scenarios for the mid- and late 21st centuries was determined, with the findings illustrated in Fig. 13.

In comparison to the baseline period, the Lake Ebinur basin is projected to experience an increase in the water supply deficit, ranging from 45 % to 113 % under future scenarios. For the mid- and late 21st century, water supply deficits were estimated for the base period and SSP scenarios (SSP126 to SSP585) as follows: 0.92 billion $m^3$ for the base period, 1.39 (1.40) billion $m^3$ for SSP126, 1.35 (1.59) billion $m^3$ for SSP245, 1.34 (1.84) billion $m^3$ for SSP370, and 1.41 (1.96) billion $m^3$ for SSP585, respectively. Values for the mid and late 21st century are shown in parentheses. Future water supply deficits are projected to become more severe under various scenarios, with higher-intensity human activities associated with carbon emissions expected to result in larger water supply deficits. The scenario with the highest intensity of human activities, SSP585, is projected to reach a deficit of 1.96 billion $m^3$ in the late 21st century. Meanwhile, SSP245, which is more aligned with sustainable development, is seen as a relatively sustainable option. In this scenario, the water supply deficit was projected to reach 1.35 billion $m^3$ in the mid-21st century, increasing to 1.59 billion $m^3$ in the later period. The findings of this study are expected to provide scientifically valuable guidance for the optimal allocation and efficient use of water resources in arid inland regions.

### 4.3. Strategies for Adapting to future water consumption challenges

To address the projected water deficits ranging from 45 % to 113 % under future SSP scenarios, targeted strategies are required to achieve sustainable water management in the Lake Ebinur basin. Improving agricultural water use efficiency is essential, as agriculture accounts for 80 % of total water consumption (Han and Jia, 2022). Optimal irrigation water allocation methods, such as multilayer multi-objective cooperative systems, have demonstrated potential reductions in water use by up to 17 % while decreasing the agricultural water use ratio by 24 % (Li et al., 2023). These methods ensure fairness of water distribution and control allocation errors within 1 %, contributing to high economic benefits. Promoting advanced irrigation technologies and drought-tolerant crop varieties can further optimise water usage and alleviate projected water deficits of up to 1.96 billion $m^3$ under SSP585. Restoring ecological systems is equally critical, given the projected 49.11 % increase in ecological water consumption under SSP585. Ecological restoration programmes, such as afforestation, are often implemented to enhance vegetation cover and soil retention. Large-scale greening can lead to substantial increases in evapotranspiration, further exacerbating

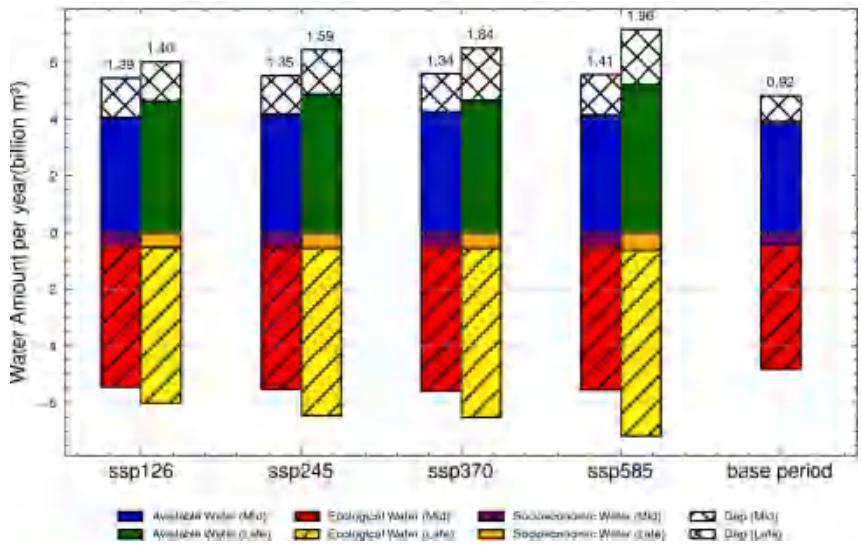


**Fig. 13.** Bar charts of water supply and consumption deficits for the Lake Ebinur Basin under different SSP scenarios during the mid- to late 21st century. (Red, yellow, green, and blue bars represent mid-21st century ecological water consumption, late-21st century ecological water consumption, late-21st century available water resources, and mid-21st century available water resources, respectively. Orange and purple bars correspond to late-21st century socio-economic water consumption and mid-21st century socio-economic water consumption, respectively. White diagonal hatching indicates water deficits under different SSP scenarios and the baseline period, with specific deficit values labeled on the chart.).

terrestrial water storage depletion, as observed in the North China Region (NCR), with a decline of − 8.9 mm/yr ($p < 0.01$) (Zhou et al., 2023). This has highlighted the importance of carefully managing ecological restoration efforts to ensure water balance, particularly in arid regions where excessive water loss from vegetation may undermine sustainable water resource management.

Strengthening water governance and exploring inter-basin water transfers have been proposed as long-term solutions for addressing water deficits. Transferring surplus water from the Kashgar River in the Ili River Basin has been suggested to offset deficits exceeding 1.35 billion $m^3$ under SSP245, with minimal impacts on donor regions. Simultaneously, the implementation of real-time water monitoring systems, policy-driven water allocation, and incentives for water-saving practices could further optimise resource distribution and enhance regional water resilience. By integrating these strategies, escalating water consumption and water deficits in the Lake Ebinur basin can be effectively mitigated, providing a suitable framework for sustainable water management in inland arid regions.

## 5. Conclusions

To the best of our knowledge, no literature has been published on the use of physical hydrological models combined with the latest CMIP6 output data to assess future agricultural and ecological water consumption and water scarcity in arid regions. Most previous studies have focused on the impacts of climate change on water availability and crop productivity. We aimed to fill this gap by predicting spatiotemporal changes in agricultural and ecological water consumption and water supply deficits under different SSP pathways. In this study, a framework was proposed to assess the future water consumption of agriculture and ecosystems and water supply deficits under various SSP scenarios. The framework consists of three main steps: downscaling the CMIP6 climate model output data, modelling with the VIC hydrological model and uncertainty analysis, and assessing the spatiotemporal changes in agricultural and ecological water consumption and water supply deficits.

After the application of QDM bias correction, the discrepancies between the observed and model-derived values of precipitation, temperature (maximum and minimum), and wind speed were significantly reduced, with differences decreasing from 45.23 mm, −1.60 °C, −4.00 °C, and 1.13 m/s to − 1.32 mm, −0.01 °C, 0.003 °C, and 0.002 m/s, respectively, demonstrating its effectiveness. Substantial improvements in runoff and evapotranspiration simulation accuracy were achieved with the bias-corrected VIC model, driven by downscaled CMIP6 data. Runoff simulations showed an increase of approximately 130 % and 128 % during the calibration and validation periods, respectively. Evapotranspiration accuracy improved significantly, as evidenced by $R^2$, NSE, and RMSE values of 0.87, 0.86, and 48.40 mm, reflecting an 86.79 % enhancement over pre-bias correction figures. Under the SSP126 to SSP585 scenarios, projections for agricultural water consumption indicated an increase of 8.00 % to 9.02 % in the mid-21st century, with further increases of 21.43 % to 43.13 % by the late 21st century. Meanwhile, ecological water consumption is expected to increase by 13.31 % to 15.86 % in the mid-21st century, with further increases ranging from 25.18 % to 49.11 % by the end of the century. Concerning adaptation strategies for future water consumption, optimising irrigation, promoting water-efficient crop varieties, and implementing inter-basin water transfer projects were essential to mitigate the projected water supply deficits. These ranged from 45 % to 113 % compared to the baseline period, with the SSP245 scenario – being the most sustainable and optimal development pathway – showing deficits of 1.35 billion $m^3$ (midterm) and 1.59 billion $m^3$ (late).

The findings of this study provide valuable insights into efficient water resource use and sustainable development in arid regions, and a novel technical framework for estimating water consumption and deficits has been proposed.

## CRediT authorship contribution statement

**Qingling Bao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jianli Ding:** Supervision, Resources, Funding acquisition. **Jinjie Wang:** Project administration. **Lijing Han:** Funding acquisition. **Jiao Tan:** Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2025.110083.

## Data availability

Data will be made available on request.

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